User Lab Platform Setup

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Overview

In this document, we first will introduce how to create a user lab platform to perform activities for our Eureka labs. Next, we will exercise with a few important wireless related commands. Lastly, we will perform a rudimentary wireless network survey with these tools and commands.

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1 Creating a Bootable Ubuntu USB

First of all, we need to set up and configure a comprehensive suite of networking and "hacking" tools to perform our labs. A >=32 GB USB flash drive is **required** for the following procedures.

Download a Ubuntu Image: For this class, we will use Ubuntu 20.04. Download the image from `http://csfaculty.tcu.edu/ma/cosc40033/tools/Linux/ubuntu-20.04-desktop-amd64.iso`. This is a large file, and the download will take time to complete.

1.1 Flash USB Drive with Rufus on Windows

You will download an image writer named Rufus from `https://rufus.ie/` and install it on your computer.

Plug in your USB drive and open Rufus on your computer. First, select the appropriate USB device. Then, select the Ubuntu Linux .ISO image downloaded earlier. Next, choose persist storage size to be 15GB to 20GB. Choose other options as shown in Fig. 1. Finally, click "START". The Ubuntu Linux image will be written to your chosen USB device.

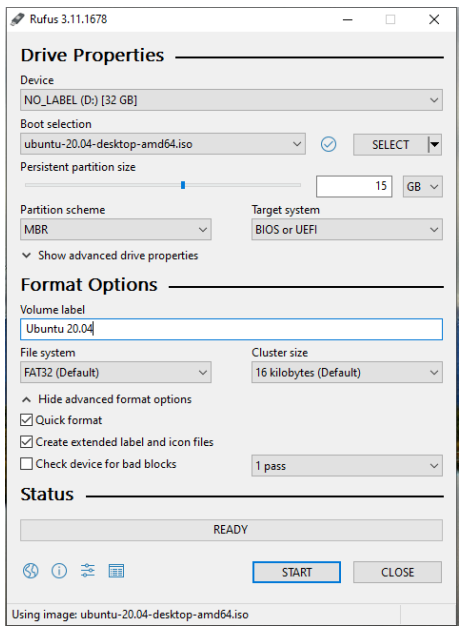


Figure 1: Rufus options.

This is the easiest option to create a Ubuntu Live USB with persistent storage so far.

1.2 Flash USB Drive on MacOS

There is **NOT** an easy solution to create a Ubuntu Live USB drive with persistent storage on Mac. A few less ideal solutions are provided below.

Option 1: Flash through a Windows Virtual Machine. Create a Windows 10 or 7 virtual machine using VMware Fusion (or VirtualBox). On the Windows virtual machine, create a Ubuntu

Live USB drive with persistent storage according to the steps listed in Section 1.1.

Option 2: Flash with Etcher without persistent storage. By default, this option will not save files or command histories; it only allows you to use the operating system. If you store your files online (OneDrive, Box, etc.) or offload them to another USB drive, this option may be fine.

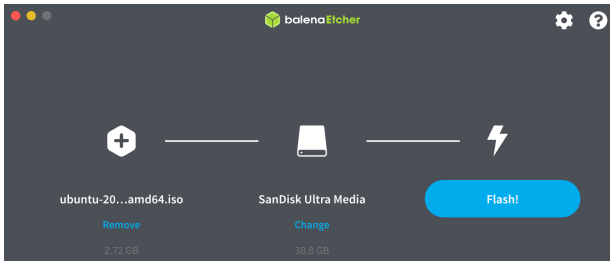


Figure 2: Etcher options.

Download Ether from `https://www.balena.io/etcher/` and install it on your Mac. Plug in your USB drive and open Etcher on your computer. First, select the Ubuntu Linux .ISO image downloaded earlier. Then, select the appropriate USB device and click "Flash!" as shown in Fig. 2. The Ubuntu Linux image will be written to your chosen USB device.

Option 3: Flash with mkusb on Ubuntu. This option is pretty complicated. Reference `https://help.ubuntu.com/community/mkusb`.

1.3 Boot from USB Sticks

1.3.1 PC

Almost all modern PCs can boot from USB sticks. However, how you tell the system to boot from a USB stick varies substantially from system to system. Initially, you can try this:

- Power off the computer.
- Plug the USB drive into a USB port.
- Power on the computer.

If the computer is configured to automatically boot from the USB drive, you will see a screen that says "Automatic boot in ..." with a countdown. If you do a native UEFI boot, where you will see a rather more minimal boot menu.

If the computer starts to boot off the hard drive as normal, you will need to manually configure it to boot off the USB drive. Usually, that should work like this:

- Reboot the computer.

- As the machine starts to reboot, watch carefully for instructions on which key to press. Usually a function key, Escape, Tab or Delete is to be pressed to enter the boot device selection menu, BIOS setup, firmware, or UEFI. Press and hold that key. If you miss the window of opportunity, often only a few seconds, then reboot and try again.
- Use the firmware, BIOS, interface or the boot device menu to put your USB drive first in the boot sequence. It might be listed as a hard drive rather than a removable drive. Each hardware manufacturer has a slightly different method for doing so.

1.3.2 Mac

Follow the steps listed below to get your Mac to load from a USB drive.

- Insert the USB boot media into an open USB slot.
- Press the Power button to turn on your Mac (or Restart your Mac if it's already on).
- When you hear the startup chime, press and hold the **Option** key. Holding that key gives you access to OS X's Startup Manager. Once the Startup Manager screen appears, release the Option key. The utility will look for any available drives that include bootable content.
- Use the arrow keys on the keyboard, select the "EFI Drive" you wish to boot from. Once selected, either hit the Return key. The machine will start to boot from the USB drive.

If everything works correctly, you should see the Ubuntu Linux setup screen on your computer. All Live USB boots activate the root user and do not require a login.

Note: after booting the USB to a Windows-based computer, Live USB's may not function correctly on MacOS devices.

For newer Mac laptops with the Apple T2 Security Chip, the Startup Security Utility prevents it booting from an external drive. Please modify Startup Security Utility according to `https://support.apple.com/en-us/HT208198`.

1.4 Install Software Tools on Live USB

Boot into your Ubuntu USB. Connect to the Internet (through WiFi or wired).

Open a terminal, and issue the following command

```
sudo add-apt-repository universe
```

to enable the "universe" source for software installation. Reference: `https://linuxconfig.org/how-to-enable-disable-universe-multiverse-and-restricted-repository-on-ubuntu-20-04-lts-focal`

Next, we will be able to install Aircrack-ng and Wireshark.

```
sudo apt-get update
sudo apt-get install aircrack-ng wireshark
```

2 Software Tools for MacOS

There are free software tools for MacOS that can help survey wireless networks, and capture and analyze wireless traffic.

- WiFi Explorer Lite allows you to survey wireless networks in vicinity using the built-in wireless NIC. This tool has a graphical user interface (GUI).
- Airtool that can capture frames with radiotap headers using the built-in wireless NIC. This tool can be used in future labs to capture wireless data frames.
- Wireshark is a widely-used network protocol analyzer.

Note that these tools can complete certain tasks, but cannot replace aircack-ng. Thus, you will still need to have a Ubuntu/Kali Linux on USB.

3 Creating a Bootable Kali USB

3.1 Download a Kali Image

For best performance, download the full "Kali Linux 64-bit" image available from `https://www.kali.org/downloads/`. This is a large file, and the download will take time to complete. Other, smaller images may work just as well for your needs, but we recommend the fully-fledged one.

3.2 Flash USB Drive with Etcher

Plug in your USB drive and open Etcher on your computer. First, select the Kali Linux .ISO image downloaded earlier. Then, select the appropriate USB device and click "Flash." The Kali Linux image will be written to your chosen USB device. This similar to what is shown in Fig. 2.

3.3 Add Persistent Storage

In this example, we assume:

- You are running as the root user. As of Kali Linux 2020.1, root user are not used by default. You can switch to root using the command "sudo bash".
- Your USB drive is **/dev/sda**. You can check it with the **lsblk** command.
- We will be creating a new partition (taking the empty space on the USB drive) to store our persistent data in.

Next, follow these steps to add persistent storage (reference: `https://www.kali.org/docs/usb/kali-linux-live-usb-persistence/`).

1. Boot into Kali Linux "Live USB Persistence" mode.
2. Open a terminal and make a new partition for persistence storage with the parted command as follows (again, assume that the USB drive appears as /dev/sda):
   `# parted /dev/sda`
3. Run `print` inside parted. Take note of where the last (second, /dev/sda2) partition ends. Assume that it ends at 3484MB.
4. Make a new partition: `mkpart primary 3484 30000` for a 32 GB USB whose second partition ends at 3484 (I left room at the tail end).
5. Quit `parted` (`quit`).
6. Make an EXT3 file system on the new partition:
   `# mkfs.ext3 /dev/sda3`
7. Label it as "persistence":
   `# e2label /dev/sda3 persistence`.
8. Mount your new partition as follows:
   ```
   # mkdir -p /mnt/usb
   # mount /dev/sda3 /mnt/usb
   # echo "/ union" > /mnt/usb/persistence.conf
   # umount /dev/sda3
   ```
9. Reboot into Live USB Persistence.
10. Open a terminal and run
    `# touch test` to create a file in the home directory of the user kali.
11. Reboot into Live USB Persistence and verify that test is still in the directory.

Congratulations! Your persistent Live USB is ready to go. Note that you need to select the "Live USB Persistence" option at each boot.

For Kali Linux, the aforementioned software tools have been installed. When using Kali Linux, most of the commands are similar to the ones used for Ubuntu. Yet, you will still need to do some "translation".

4 Wireless Related Commands

The primary purpose of this task is to help you get familiar with wireless related commands on Linux. Additionally, you can build a reference manual for your future labs.

4.1 ip

The ip command is used to assign an address to a network interface (NIC) and/or configure network interface parameters on Linux operating systems. This command replaces old good and now deprecated ifconfig command on modern Linux distributions.

The syntax for the ip command is as follows:

```
ip [ OPTIONS ] OBJECT { COMMAND | help }
```

OBJECT is the object type that you want to manage. The most frequently used object (or subcommands) in this class is **link**, which can display and modify network interfaces.

For example, to show all the available NICs on a computer, issue the following command.

```
sudo ip link show
# If you first do ``sudo bash'' to become root,
# you will no longer need to use sudo
```

If you want to list a specific NIC, you will need to specify the NIC's name in the end.

```
sudo ip link show <nic-name>
# Replace nic-name with the actual NIC device name
```

To power off/up (like disable and enable) a NIC, use the following commands:

```
sudo ip link set <nic-name> down
sudo ip link set <nic-name> up
```

This is important because certain options (e.g., change the operation mode) cannot be set without turning the NIC down first.

Here is an ip command cheat sheet for your reference.

4.1.1 Provide a screenshot of the "sudo ip link show" command results on your Ubuntu Linux.

4.1.2 Shut down and power on your wireless NIC; And provide screenshots.

4.2 iwconfig

The *iwconfig* command is similar to *ifconfig*, **but is dedicated to wireless networking interfaces**. It can be used to set the parameters of the network interface which are specific to the wireless operation (e.g., frequency, network name/SSID, and mode).

Open a terminal, and issue the following command to identify available wireless NICs and their information/parameters on your computer. An example of screenshot is given in Fig. 3, where the wireless NIC device name is "wlp108s0".

```
sudo iwconfig
```

If you want get the specific information about a NIC device, you can issue the following command.

```
sudo iwconfig <nic-name>
# In our example, it would be:
# sudo iwconfig wlp108s0
```

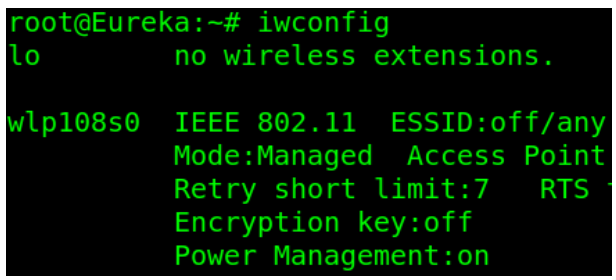


Figure 3: iwconfig command example.

To set a NIC to a specific channel (say 10):

```
sudo iwconfig <nic-name> channel 10
# Since channel 10 is 2.457G, you can also
sudo iwconfig <nic-name> freq 2.457G
```

Monitor mode is a data capture mode that allows using a WiFi adapter in listening mode or promiscuous mode. Operating in this mode, WiFi network cards are able to capture all types of WiFi Management packets (including Beacon packets), Data packets and Control packets. To set a NIC in the monitor mode, use the the following command.

```
sudo iwconfig <nic-name> mode monitor
```

Note that this command may not work for some Linux distributions and/or certain scenarios.

More example iwconfig commands can be found in `https://www.geeksforgeeks.org/iwconfig-command-in-lin`

4.2.1 Write down a specific iwconfig command to set your NIC to channel 6; And provide a screenshot.

4.2.2 Write down a specific iwconfig command to connect your NIC to a network named "eduroam"; And provide a screenshot.

4.3 iw

The *iw* command is a newer command which is more powerful than iwconfig, but different syntax that iwconfig.

To show the specific information about a NIC, issue the following command.

```
sudo iw <nic-name> info
```

You may want to use **"sudo iw dev" or "sudo iw list"** first to get the wireless NIC name.

To scan for wireless iWIFi) networks in vicinity, use the scan option as follows.

```
sudo iw <nic-name> scan
```

Set the NIC to a specific channel (say 10):

```
sudo iw <nic-name> set channel 10
# Since channel 10 is 2.457G, you can also
sudo iw <nic-name> set freq 2.457G
```

Note that sometimes the iw command may not be able to set a channel for various reasons. You can and should try the iwconfig command. In case you want to monitor 5GHz (e.g., 802.11n) traffic, you will need to specify channel width (20 or 20/40MHz) and in case of 20/40MHz if the upper or lower channel is being used (set channel 36 HT20/HT40-/HT40+).

More examples cab be seen at `https://wireless.wiki.kernel.org/en/users/documentation/iw`.

4.3.1 Write down a specific iw command to connect your NIC to a network named "eduroam"; And provide a screen shot.

4.3.2 Write down a specific iw command to set your NIC to the monitor mode; And provide a screen shot.

A hint:

```
sudo iw <nic-name> set monitor none
```

5 Wireless Network Survey

Boot into your Ubuntu USB. You can the built-in wireless network interface card (NIC) or a USB wireless NIC provided by the instructor to perform wireless survey activities.

Note that, for MAC users, you could use WiFi Explorer Lite to survey wireless networks in vicinity using the built-in wireless NIC as well (thus skipping Subsection 5.1 and the airodump-ng command in Subsection 5.2).

5.1 Driving Wireless NIC into Monitor Mode

Open a terminal, and issue the following command

```
sudo iwconfig
Or sudo iw dev
# If you do ``sudo bash'' to become root,
# you will no longer need to use sudo
```

to identify the available wireless NICs. An example of screenshot is given in Fig. 3, where the wireless NIC is named "wlp108s0". We will use this name for the subsequent commands. .

Next, we will drive the NIC card into the monitor mode so as to survey the wireless networks in vicinity. Instead of using iwconfig or iw, we will use an option (via the *airmon-ng* command) that has the best compatibility.

```
sudo airmon-ng check kill
# This will stop any process that
# prevents the card into the monitor mode

sudo airmon-ng start wlp108s0
# This would drive the NIC named
# wlp108s0 into the monitor mode.
# You will need to replace it with
# the actual NIC name on your computer
```

An example of screenshot is given in Fig. 4. Note that the NIC name becomes "wlp108s0mon", which indicates that the NIC card is in the monitor mode and ready to collect wireless frames.

5.1.1 When using airmon-ng to drive a NIC to the monitor mode, how to fix the NIC to a specific channel?

5.2 Surveying Wireless Networks in Vicinity

Issue the following command to start survey the wireless networking near you.

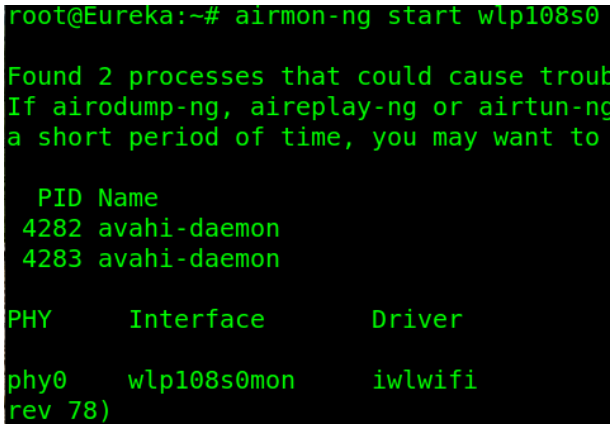


Figure 4: airmon-ng start wlp108s0 results.

```
sudo airodump-ng wlp108s0mon
# This starts survey on wlp108s0mon
# by sweeping all available channels
```

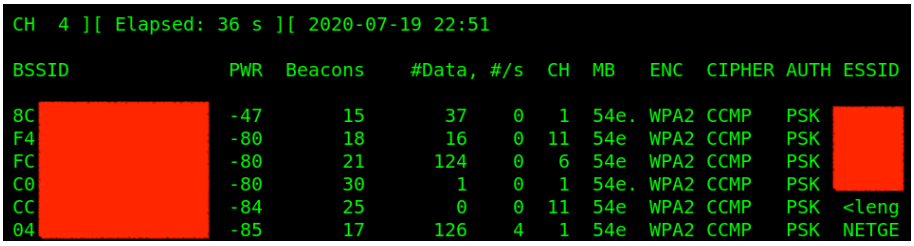


Figure 5: airodump-ng wlp108s0mon results.

An example screenshot is shown in Fig. 5. Some portion of the figure is redacted for privacy reasons. Next I will explain some terminologies shown in the figure. BSSID means the medium access (MAC) address of the access point (AP). PWR refers to the power strength of the received wireless signals. CH means the channel that the wireless network is operating at. MB refers to the speed of the wireless network. ENC, CIPHER, and AUTH refer to the security protocols adopted by the wireless network. ESSID refers to the wireless network name.

5.2.1 Take a screenshot of your wireless survey results.

5.2.2 When using airodump-ng, how to fix the NIC to survey on a specific channel?

5.2.3 Why does the EESID shown as "<leng" (would be "<length: 0>" if captured in full) instead of its real network name in Fig. 5? Did you see that in your survey?

In a later section, you will be able to practice this technique (SSID hiding).

6 Flashing Router with OpenWRT

Each wireless router has a certain version of firmware. You will need to flash your router (Linksys WRT3200ACM) to the OpenWRT firmware, which is a Linux operating system targeting embedded devices. OpenWrt provides a fully writable filesystem with package management. It allows you to customize the device through the use of packages to suit any application. For developers, OpenWrt is the framework to build an application without having to build a complete firmware around it; fFor users this means the ability for full customization, to use the device in ways never envisioned. Reference: `https://openwrt.org/`.

6.1 Flash Router with OpenWRT via WebUI

The steps listed below for flashing a Linksys WRT3200ACM are adopted from: `https://openwrt.org/toh/linksys/linksys_wrt3200acm`.

1. Download the OpenWRT firmware for his class to your computer from: `http://csfaculty.tcu.edu/ma/cosc40033/tools/Linksys-AC3200/Firmware/OpenWRT/openwrt-19.07.3-mvebu-cortexa9-wrt3200acm-squashfs-factory.img`.
2. Power on your router.
3. Connect your computer to one of the LAN ports on the router using a network cable (wired connection). Please do NOT connect to the WAN port.
4. Open a browser and type in the router's Default IP: 192.168.1.1. Choose manual configuration for the first time login.
5. Login with the default Linksys router password: admin.
6. Go to "Connectivity" → "Basic" → "Choose File" (as shown in Fig. 6) and choose the firmware downloaded in the previous step. Next, click "Start".
7. Once flash completes, router will reboot.

6.2 Set Root Password and Enable SSH

After the router restarts, you can log onto the router via a browser with the IP address of 192.168.1.1. After logging in, you will need to configure a root password to protect the web interface and enable SSH as shown in Fig. 7b.

Next, try login to the router via SSH. On a terminal of your computer, ssh to the router (192.168.1.1) using the following command:

```
ssh  root@192.168.1.1
```

and type your password. If successful, you will see a screen like Fig. 7c.

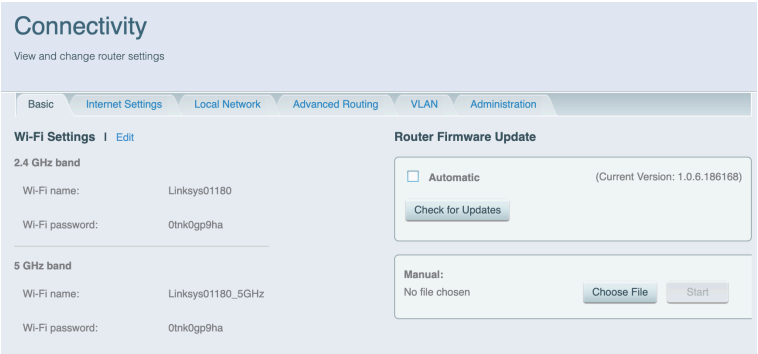
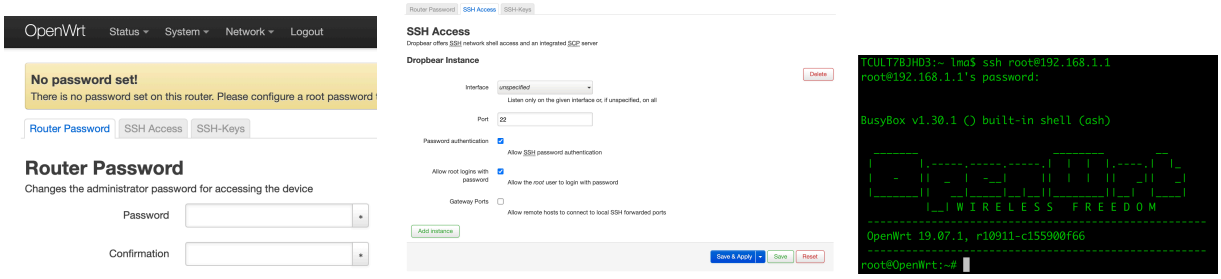


Figure 6: Flash Linksys with OpenWRT.



(a) Set root password. (b) Enable SSH. (c) SSH login.

Figure 7: Configure a root password and enable SSH.